

ENG4200 Coursework 1 Report

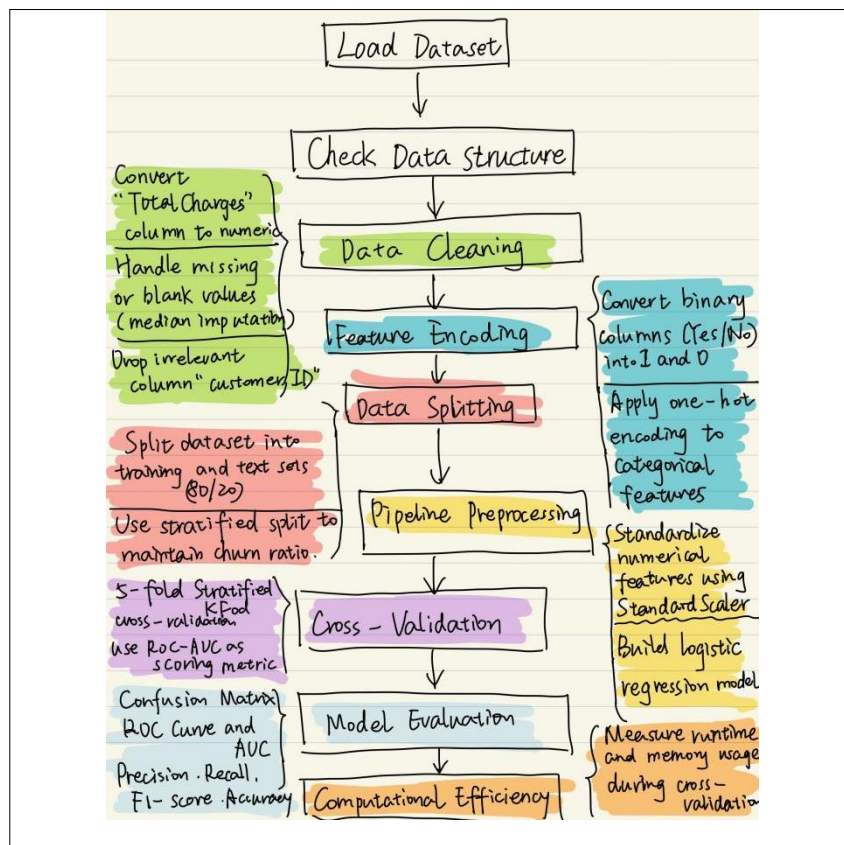
Dataset: Telco Customer Churn

1. Objective (3–4 sentences)

The objective of this coursework is to develop and evaluate a binary classification model to predict customer churn in the Telco Customer Churn dataset. Using scikit-learn, a machine learning pipeline was achieved to integrate data preprocessing and model training. The aim of this coursework is to create a binary classification model to capture patterns that distinguish customers likely to leave from who remain and assess how effectively.

2. Pipeline Design

Provide a schematic/flow diagram of your pipeline. (Hand-drawn or software generated, include as a figure.)



3. Key Results

- Include one plot or table with your main performance metrics.

	precision	recall	f1-score	support
0	0.851	0.894	0.872	1035
1	0.658	0.567	0.609	374
accuracy			0.807	1409
macro avg	0.755	0.730	0.741	1409
weighted avg	0.800	0.807	0.802	1409

- Summarise accuracy, efficiency, or loss curves concisely.

During 5-fold cross-validation, the model achieved a mean ROC-AUC score of approximately 0.846, indicating good discrimination between churned and non-churned customers. On the test set, the Logistic Regression pipeline achieved the following performance metrics:

Accuracy: 0.807

Precision (Churn=1): 0.658

Recall (Churn=1): 0.567

F1-score (Churn=1): 0.609

ROC-AUC: 0.842

These results demonstrate reasonable predictive performance, but the model shows a tendency to miss some positive (churn) cases due to lower recall.

The model training and cross-validation completed in approximately 0.13 seconds with 5.61 MB of additional memory usage.

4. Analysis & Reflection

- What worked well?

The overall pipeline structure worked very well. Integrating data preprocessing and model training into a single scikit-learn pipeline kept the workflow clean, organized, and fully reproducible. Using StandardScaler to standardize numerical features significantly improved the stability of model training and allowed the Logistic Regression algorithm to converge more efficiently. The use of StratifiedKFold cross

-validation provided a reliable estimate of model performance by maintaining balanced class proportions across folds and reducing the influence of data imbalance. Additionally, the pipeline design minimized the risk of data leakage, since scaling and fitting occurred within the same structure during each validation split.

- **What difficulties or limitations did you encounter?**

One main difficulty was dealing with the class imbalance in the dataset, where only around one quarter of customers actually churned. This imbalance caused the model to favor the majority class, resulting in lower recall for churned customers. Another limitation came from the linear nature of Logistic Regression. Some customer behaviors are likely influenced by nonlinear or interacting factors that a linear model cannot capture well. Additionally, several categorical features were highly correlated, such as contract type and tenure, which may have reduced the model's ability to learn distinct patterns. Although cross-validation helped improve reliability, there was still some variation in scores across folds, suggesting that certain subsets of the data were more difficult to model due to noise or limited sample size.

- **How did you address (or attempt to address) overfitting?**

Overfitting was addressed through the use of cross-validation and model regularization. A 5-fold StratifiedKFold cross-validation ensured that the model was trained and validated on multiple data subsets, reducing the risk of learning noise from a single split. The training and validation ROC-AUC scores suggest that no serious overfitting occurred.

5. Next Steps (2–3 sentences)

If you had more time, how would you improve or extend the work?

A recall of 0.567 for churned customers implies that the model failed to capture some minority-class patterns, likely due to class imbalance. If more time were available, I would focus on improving the model's recall for churned customers by addressing class imbalance and optimizing decision thresholds. Techniques such as class weighting or oversampling to help the classifier better identify minority cases.

Appendix (optional, max 1 page)

Extra plots/tables if strictly necessary. Main analysis should stay concise in Sections 1–5.

